

URBANNEST HOSPITALITY ANALYTICS

Hotel Revenue Prediction — Capstone Project Report

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Dataset: Hotel Booking Demand Dataset (2015-2017)

Model: Optimised Linear Regression

1. Executive Summary

This report presents the findings of a hospitality analytics engagement conducted for UrbanNest Hospitality Analytics. The project involved a comprehensive analysis of hotel booking data spanning July 2015 to August 2017, encompassing 119,390 raw booking records across two hotel types — City Hotel and Resort Hotel.

Following a rigorous data cleaning pipeline, 77,678 valid booking records were retained for analysis. The primary objective was to build a predictive model for hotel booking revenue using key booking and customer features. Following exploratory data analysis and feature engineering, an optimised Linear Regression model was developed achieving an R^2 score of 0.7844 and a Mean Absolute Error of \$86.63 per booking.

Key findings include: Online TA dominates booking volume but also carries the highest cancellation risk; room assignment changes reduce cancellation rates by 26 percentage points; and total nights stayed is the single strongest predictor of booking revenue. These insights provide UrbanNest with actionable intelligence for improving pricing, occupancy, and revenue performance across their hotel client portfolio.

2. Business Problem Statement

The hospitality industry is highly competitive, and small differences in pricing, demand, and customer behaviour can significantly affect revenue. Many hotels continue to rely on intuition or historical assumptions when making pricing and revenue decisions, often leading to missed revenue opportunities during peak periods and underperformance during low-demand seasons.

UrbanNest Hospitality Analytics contracted this data science team to address the following business question:

"What factors drive hotel booking revenue, and can we build a reliable model to predict booking value based on guest and booking characteristics?"

Specifically, the engagement sought to:

- Identify the key drivers of hotel booking revenue
- Understand booking patterns, customer behaviour, and cancellation risk
- Build a predictive model for revenue forecasting
- Provide actionable recommendations to improve pricing and occupancy decisions

Better revenue predictions enable hotels to optimise room pricing dynamically, reduce cancellation-related revenue leakage, and identify high-value customer segments worth targeting.

3. Dataset Description

The Hotel Booking Demand Dataset was used for this analysis. It contains booking information for two hotel types — City Hotel and Resort Hotel — and includes features such as booking duration, number of guests, room type, lead time, seasonality, and revenue indicators.

Attribute	Details
Source	Hotel Booking Demand Dataset (publicly available)
Time Period	July 2015 to August 2017
Raw Records	119,390 bookings
Clean Records	77,678 bookings
Features	33 original columns
Hotel Types	City Hotel and Resort Hotel
Target Variable	Total Revenue (ADR × Total Nights)

Note: 2015 and 2017 represent partial years (July-August 2015 and January-August 2017 respectively). Year-level comparisons should be interpreted with caution.

4. Data Cleaning & Preprocessing

A comprehensive data cleaning pipeline was implemented to ensure data quality and reliability for modelling. The following decisions were made and justified:

4.1 Missing Values

Column	Missing %	Action	Justification
company	94.3%	Dropped	Insufficient data for meaningful analysis
country	0.4%	Rows dropped	Only 488 rows — negligible loss
agent	13.7%	Filled with 0	0 represents direct bookings with no agent
children	0.003%	Filled with 0	Only 4 nulls — clearly no children

4.2 Data Inconsistencies

- 31,933 duplicate rows removed — retaining one record per unique booking
- 715 zero-night stay rows removed — consistent with day-use bookings outside project scope
- 180 ghost bookings removed — bookings with zero adults, children, and babies
- arrival_date_month converted to ordered categorical for correct chronological sorting
- reservation_status_date converted from string to datetime
- Undefined values in market_segment, distribution_channel, and meal replaced with modal category

4.3 Outlier Detection

The IQR method was selected for outlier detection as key columns exhibited positive skewness, making it more robust than the standard deviation approach which assumes normality. Outlier rows were removed rather than capped to preserve data integrity.

Column	IQR Upper Bound	Action
adr	\$211.06	Rows outside bounds removed
lead_time	373 days	Rows outside bounds removed
stays_in_week_nights	8 nights	Rows outside bounds removed
stays_in_weekend_nights	5 nights	Rows outside bounds removed

4.4 Feature Engineering

- Zero ADR rows (1,959) removed — represent complimentary stays generating no revenue
- total_revenue derived as $ADR \times \text{total nights stayed}$ — captures full booking value
- room_changes binary feature created — flags bookings where assigned room differed from reserved room
- total_nights created by combining stays_in_week_nights and stays_in_weekend_nights

5. Exploratory Data Analysis

Extensive EDA was conducted across eight analytical sections to understand booking patterns, customer behaviour, and revenue drivers.

5.1 Univariate Analysis

- ADR is approximately normally distributed with a mean of \$101.83 and median of \$94.58, with mild positive skew from premium bookings
- Total revenue per booking is right-skewed (mean \$348.69, median \$288.00) — a small number of high-value bookings pull the average upward
- City Hotel accounts for 63.7% of bookings vs Resort Hotel at 36.3%
- 26.8% of bookings were cancelled following data cleaning — representing significant revenue risk

5.2 Time & Seasonality

- Peak booking months are July and August (8,500-8,900 bookings), with a secondary peak from March to June
- January and November-December are the lowest volume months (4,400-4,700 bookings)
- Revenue peaks in August ($\approx \$4.7\text{M}$ aggregate) and drops sharply in winter months
- 2016 was the only complete year in the dataset — partial years 2015 and 2017 should not be compared directly

5.3 Cancellation Analysis

- Online TA has the highest cancellation rate at 34.2% — concerning given it also dominates booking volume
- Cancellation rate increases consistently with lead time: 18.3% for 0-30 days vs 38.8% for 181-300 days
- Transient customers cancel at 29.2% vs Group customers at just 6.3%
- Non-refundable deposit bookings show a counterintuitive 92.8% cancellation rate, likely reflecting data recording conventions

5.4 Revenue Analysis

- Online TA generates the highest average revenue per booking at \$384, followed by Direct at \$336
- Contract customers generate the highest average revenue at \$409 with low cancellation rates — the most valuable customer type
- Room C generates the highest average total revenue at ~\$630 per booking despite not having the highest ADR
- Average revenue increases consistently with number of special requests — from \$320 (0 requests) to \$510 (5 requests)

5.5 Key Insight — Room Changes

The most actionable insight in the dataset: bookings where the room was changed show a cancellation rate of just 4.8% compared to 31.6% for unchanged bookings. Guests receiving a room upgrade are nearly 7 times less likely to cancel, suggesting proactive room reassignment is an effective revenue protection strategy.

5.6 Correlation Analysis

The correlation matrix identified the following key relationships with total revenue:

Feature	Correlation with Total Revenue	Interpretation
stays_in_week_nights	0.70	Strongest predictor — duration drives revenue
adr	0.59	Nightly rate is important but secondary to duration
stays_in_weekend_nights	0.52	Weekend stays also significant
adults	0.29	More guests correlates with higher value bookings
lead_time	0.29	Early planners tend toward longer stays
children	0.19	Family bookings show moderate revenue signal

6. Model Development & Evaluation

6.1 Feature Selection

Features were selected based on correlation strength with total revenue, business relevance, and the principle of parsimony. The final feature set comprised 28 features after one-hot encoding:

- Numerical: total_nights, adults, lead_time, children, total_of_special_requests, is_canceled, room_changes, arrival_date_month
- Categorical (encoded): hotel, market_segment, customer_type, reserved_room_type, deposit_type

6.2 Train-Test Split

The dataset was split into 80% training (62,142 rows) and 20% testing (15,536 rows) using a fixed random seed of 42 for reproducibility. This ensures the model is evaluated on unseen data, simulating real-world performance on new bookings.

6.3 Baseline Model

A baseline Linear Regression model was trained on the full feature set, achieving the following performance:

Metric	Baseline Model	Interpretation
R ² Score	0.7747	Explains 77.5% of revenue variance
MAE	\$89.00	Average prediction error per booking
RMSE	\$127.51	Penalised error — sensitive to large mistakes

6.4 Model Optimisation

Two optimisation strategies were applied to improve model performance:

Feature Elimination: is_canceled was removed as it represents data leakage — cancellation status is not available at prediction time and its relationship with revenue is consequential rather than causal. Removing it caused only a negligible reduction in R² of 0.001, confirming it added no meaningful predictive signal.

Interaction Feature Engineering: Three interaction terms were created to capture combined feature effects:

- total_nights × adults — longer stays with more guests generate disproportionately higher revenue
- total_nights × special_requests — engaged guests booking longer stays represent higher booking value
- lead_time × total_nights — early planners booking extended stays exhibit distinct revenue patterns

6.5 Final Model Performance

Metric	Baseline Model	Optimised Model	Improvement
R ² Score	0.7747	0.7844	+0.0097
MAE	\$89.00	\$86.63	-\$2.37
RMSE	\$127.51	\$124.74	-\$2.77

The optimised model explains 78.4% of the variability in total booking revenue. The MAE of \$86.63 indicates that on average predictions deviate by \$86.63 from actual revenue — reasonable given the dataset average revenue of \$348.69.

Feature scaling (StandardScaler) was tested but produced identical results, confirming a known property of Ordinary Least Squares Linear Regression — it is scale invariant. The unscaled model was retained for interpretability.

6.6 Model Limitations

- The model underpredicts high-value bookings above \$800, struggling with premium stays
- A small number of negative revenue predictions were observed — not realistic but expected with Linear Regression
- The remaining 21.6% of unexplained variance reflects dynamic pricing, promotional rates, and external demand factors not captured in the dataset

7. Business Recommendations

Based on the analysis findings, the following recommendations are made for UrbanNest and their hotel clients:

7.1 Revenue Strategy

- Focus on increasing total nights stayed rather than maximising nightly rates — duration is the strongest revenue driver (correlation 0.70)
- Develop length-of-stay incentives such as discounted weekly rates to encourage longer bookings
- Target Contract customers through direct channels — they generate the highest average revenue (\$409) with low cancellation rates (17%)

7.2 Cancellation Risk Management

- Implement stricter deposit policies for bookings with lead times exceeding 90 days, where cancellation rates exceed 34%
- Prioritise retention efforts for Online TA Transient bookings — the highest risk combination (34.2% cancellation rate, highest volume)

- Deploy proactive room upgrade strategies — guests receiving room changes cancel at only 4.8% vs 31.6% for unchanged bookings

7.3 Channel Strategy

- Incentivise direct bookings through loyalty programmes or exclusive rates — direct bookings yield \$336 average revenue with no commission costs
- Develop corporate contract pipeline — Contract customers combine high revenue and low cancellation risk
- Reduce over-reliance on Online TA which dominates 60%+ of bookings — creating strategic and revenue risk

7.4 Seasonal Strategy

- Implement dynamic pricing during peak months (July-August) to maximise ADR when demand is highest
- Develop off-season promotions for November-February to stimulate demand during low-booking periods
- Focus group and corporate bookings in shoulder seasons (March-June, September-October) to maintain stable occupancy

8. Conclusion

This capstone project successfully delivered a complete data science workflow for UrbanNest Hospitality Analytics — from raw data inspection through cleaning, EDA, SQL analysis, and predictive modelling.

The optimised Linear Regression model achieved an R^2 of 0.7844 and MAE of \$86.63, providing UrbanNest with a reliable baseline for forecasting hotel booking revenue. The model identifies total nights stayed, room type, number of adults, and guest engagement as primary revenue drivers — actionable insights directly translatable into pricing and retention strategies.

For future work, it is recommended that UrbanNest considers adopting Random Forest as the production model. In benchmarking tests, Random Forest achieved an R^2 of 0.87 and MAE of \$58.19 — a 40.6% reduction in prediction error — better capturing the non-linear relationships in hotel revenue data.